Grade 10, Term 1, C4: Maximax and Maximin

Version 2

Learning Criterion

C4: Applies the Maximax and Maximin criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case outcome.

1 Maximax and Maximin

When probabilities are unavailable, decisions can be made by comparing the payoffs for each alternative under each possible state of nature.

	Maximax	**Maximin**
Focus	Best possible payoff	Worst possible payoff
Attitude	Optimistic	Conservative
First step	Find the maximum in each row	Find the minimum in each row
Final step	Choose the largest maximum	Choose the largest minimum

$$M_i = \max_j P_{ay}(A_i, S_j)$$
$$A^*_{\text{Maximax}} = \arg\max_i M_i$$

$$m_i = \min_j P_{ay}(A_i, S_j)$$
$$A^*_{\text{Maximin}} = \arg\max_i m_i$$

In every problem, choose exactly one alternative before the external state is known. The alternatives are mutually exclusive; no state probabilities are available. All entries are profits in thousand USD, with higher values better (negative entries are losses). Use the supplied payoffs directly.

Identify alternatives A_i and states S_j. Find each row maximum and minimum, then choose the largest maximum for Maximax and the largest minimum for Maximin. Complete the summary where provided.

2 Solved Example: Boat Tours

A boat operator chooses open-sea tours or harbor tours before next month's weather is known. The month will be calm or stormy. All payoffs are profits in thousand USD.

Alternative	Calm	Stormy
Open-sea tours	46	9
Harbor tours	31	23

Step 1: Identify

A_1 = Open-sea tours A_2 = Harbor tours

S_1 = Calm S_2 = Stormy

Step 2: Decide

Maximax

$$M_1 = \max\{46, 9\} = 46$$
$$M_2 = \max\{31, 23\} = 31$$
$$A^*_{\text{Maximax}} = A_1 = \text{Open-sea tours}$$

Maximin

$$m_1 = \min\{46, 9\} = 9$$
$$m_2 = \min\{31, 23\} = 23$$
$$A^*_{\text{Maximin}} = A_2 = \text{Harbor tours}$$

Step 3: Complete the table

Alternative	Max	Min
A_1	46	9
A_2	31	23
A^*_{Maximax}	A_1	
A^*_{Maximin}		A_2

	Alternative	
	A_1	A_2
Max	46	31
Min	9	23
A^*_{Maximax}	A_1	
A^*_{Maximin}		A_2

3 Guided Practice: Translation Agency

A translation agency chooses to bid abroad or serve local clients before a foreign tender round. Tender access will be open or restricted. All payoffs are profits in thousand USD.

Alternative	Open	Restricted
Bid abroad	58	7
Local clients	38	24

Step 1: Identify

A_1 = _____
A_2 = _____
S_1 = _____
S_2 = _____

Step 2: Decide

Maximax

M_1 = max{____, ____} = _____
M_2 = max{____, ____} = _____
A^*_{Maximax} = _____

Maximin

m_1 = min{____, ____} = _____
m_2 = min{____, ____} = _____
A^*_{Maximin} = _____

Step 3: Complete the table

Alternative	Max	Min
A_1	_____	_____
A_2	_____	_____
A^*_{Maximax}	_____	
A^*_{Maximin}		_____

4 Practice: Film Rights

A film distributor chooses cinema rights or streaming rights before critics review a film. Reviews will be favorable or unfavorable. All payoffs are profits in thousand USD.

Alternative	Favorable	Unfavorable
Cinema rights	51	4
Streaming rights	33	19

Step 1: Identify

A_1 = _____ A_2 = _____
S_1 = _____ S_2 = _____

Step 2: Decide

Maximax

M_1 = max{____, ____} = _____
M_2 = max{____, ____} = _____
A^*_{Maximax} = _____

Maximin

m_1 = min{____, ____} = _____
m_2 = min{____, ____} = _____
A^*_{Maximin} = _____

Step 3: Complete the table

Alternative	Max	Min
A_1	_____	_____
A_2	_____	_____
A^*_{Maximax}	_____	
A^*_{Maximin}		_____

	A_1	A_2	A^*_{Maximax}	A^*_{Maximin}
Max	_____	_____	_____	
Min	_____	_____		_____

5 Guided Practice: Orchard

An orchard owner commits the land to one use before seasonal rainfall is known: wet, normal, or dry. A fixed lease guarantees the same net profit in every state because the tenant bears all rainfall risk. All payoffs are profits in thousand USD.

	Rainfall		
Alternative	Wet	Normal	Dry
Plant berries	84	37	-16
Plant apples	63	46	21
Plant olives	43	34	27
Fixed lease	17	17	17

Step 1: Identify

A_1 = _____ A_2 = _____

A_3 = _____ A_4 = _____

S_1 = _____ S_2 = _____

S_3 = _____

Step 2: Decide

Maximax

M_1 = max{___, ___, ___} = ____

M_2 = max{___, ___, ___} = ____

M_3 = max{___, ___, ___} = ____

M_4 = max{___, ___, ___} = ____

A^*_{Maximax} = _____

Maximin

m_1 = min{___, ___, ___} = ____

m_2 = min{___, ___, ___} = ____

m_3 = min{___, ___, ___} = ____

m_4 = min{___, ___, ___} = ____

A^*_{Maximin} = _____

Step 3: Complete the table

	A_1	A_2	A_3	A_4	Decision
Max	____	____	____	____	A^*_{Maximax} = ____
Min	____	____	____	____	A^*_{Maximin} = ____

6 Independent Practice: Canal Freight

A freight operator chooses one plan before canal water levels are known: full, limited, or closed to navigation. A fixed charter pays the same net profit in every state because the client pays all operating and rerouting costs. All payoffs are profits in thousand USD.

	Canal level		
Alternative	Full	Limited	Closed
Deep barges	94	33	-19
Shallow barges	68	47	23
Road freight	49	41	31
Fixed charter	24	24	24

Step 1: Identify

A_1 = _____ A_2 = _____

A_3 = _____ A_4 = _____

S_1 = _____ S_2 = _____

S_3 = _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	94	33	-19	_____	_____
A_2	68	47	23	_____	_____
A_3	49	41	31	_____	_____
A_4	24	24	24	_____	_____

Maximax

A^*_{Maximax} = _____

Maximin

A^*_{Maximin} = _____

7 Independent Practice: Theatre

A theatre chooses one use of its hall before a rival festival announces its dates. Date overlap will be none, partial, or full. A studio lease guarantees the same net profit regardless of overlap because the tenant bears attendance risk. All payoffs are profits in thousand USD.

	Date overlap		
Alternative	None	Partial	Full
Stage musical	89	34	-12
Stage comedy	66	51	20
Host workshops	47	39	33
Studio lease	26	26	26

Step 1: Identify

A_1 = _____ A_2 = _____

A_3 = _____ A_4 = _____

S_1 = _____ S_2 = _____

S_3 = _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	89	34	-12	_____	_____
A_2	66	51	20	_____	_____
A_3	47	39	33	_____	_____
A_4	26	26	26	_____	_____

Maximax

A^*_{Maximax} = _____

Maximin

A^*_{Maximin} = _____

8 Independent Practice: Recycling

A recycling firm chooses one use of its site before scrap prices are known: high, steady, or low. A storage lease guarantees the same net profit under all scrap prices because its rent and costs are fixed. All payoffs are profits in thousand USD.

	Scrap prices		
Alternative	High	Steady	Low
Process metals	98	31	-21
Sort plastics	73	53	19
Collect paper	52	43	35
Storage lease	29	29	29

Step 1: Identify

A_1 = _____ A_2 = _____

A_3 = _____ A_4 = _____

S_1 = _____ S_2 = _____

S_3 = _____

Step 2: Complete the analysis

Alternative	S_1	S_2	S_3	Max	Min
A_1	98	31	-21	_____	_____
A_2	73	53	19	_____	_____
A_3	52	43	35	_____	_____
A_4	29	29	29	_____	_____

Maximax

A^*_{Maximax} = _____

Maximin

A^*_{Maximin} = _____